ToxPipe Evaluation

Toxpipe Evaluation Application

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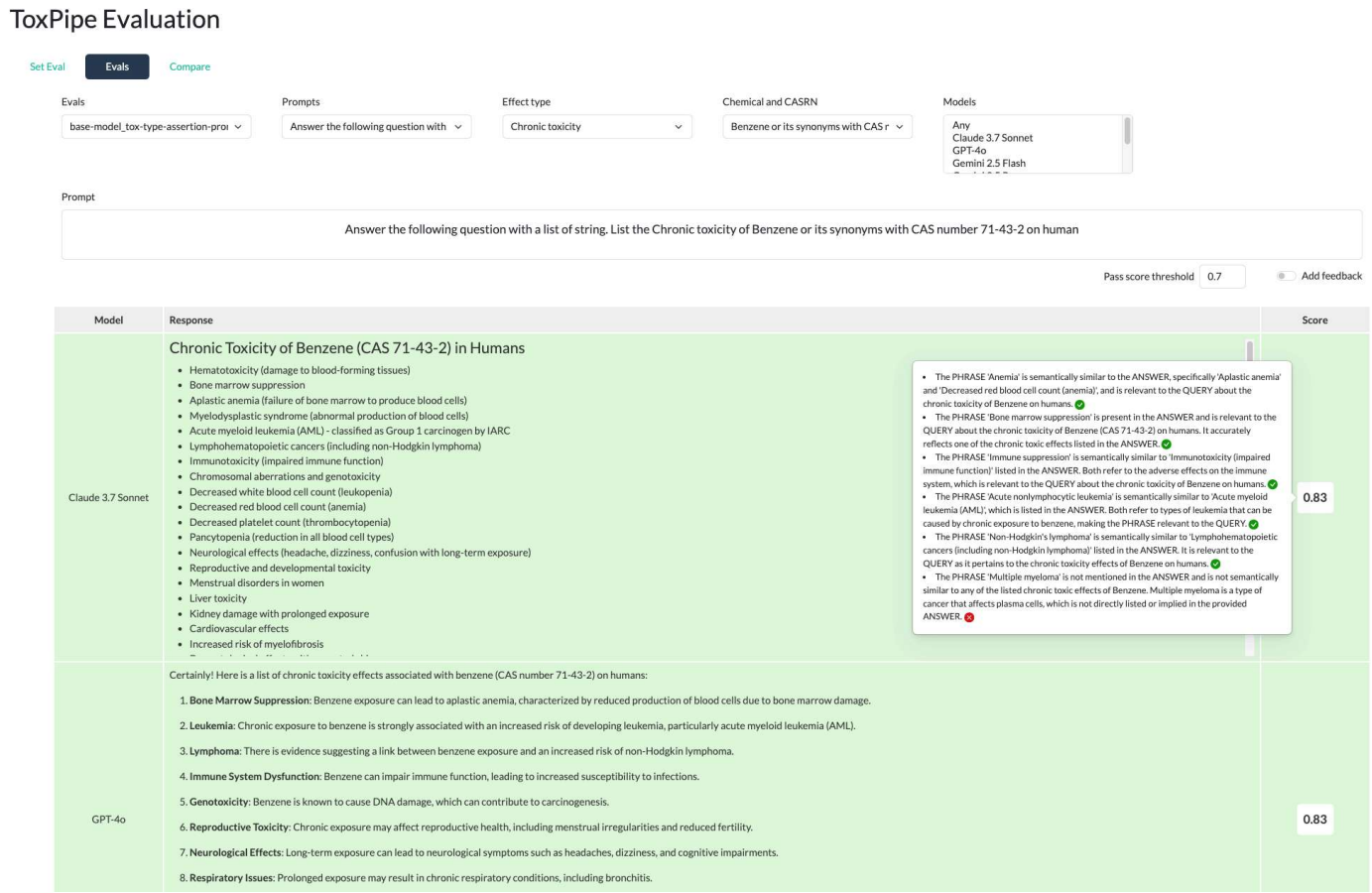
PUBLISHED
Mar 12, 2026

ToxPipe evaluation sets, evaluation results and the responses of the AI models are documented and compared in this link

AI Model responses on the evaluation sets



Comparison between responses from different AI Models

ToxPipe Evaluation

Set Eval | Evals | Compare

Prompts | Graphs | Table

Eval set: Tox type prompts (Human)

Prompts: Answer the following question with

Chemical and CASRN: Aspirin or its synonyms with CAS nu

Effect type: Sub acute toxicity

Prompt

Answer the following question with a list of string. List the Sub acute toxicity of Aspirin or its synonyms with CAS number 50-78-2 on human

Pass score threshold 1

Model	Response	Response (RAG)	Response (Agentic)
Claude 3.7 Sonnet	Sub-acute Toxicity of Aspirin (CAS 50-78-2) in Humans • Gastrointestinal irritation and erosions • Gastric ulceration and bleeding • Tinnitus (ringing in the ears) • Hearing impairment • Increased bleeding time due to platelet inhibition • Reye's syndrome (in children with viral infections) • Respiratory alkalosis followed by metabolic acidosis • Hyperthermia • Dehydration • Electrolyte imbalances (particularly potassium) • Hepatotoxicity (mild to moderate liver enzyme elevations) • Renal impairment (decreased glomerular filtration) • Hypersensitivity reactions (urticaria, angioedema) • Bronchoconstriction in susceptible individuals • Headache • Dizziness • Confusion • Hyperventilation • Nausea and vomiting	*[The following response was taken from model's training knowledge]* Sub-acute Toxicity of Aspirin (CAS 50-78-2) in Humans Based on toxicological data, the sub-acute toxicity effects of aspirin (acetylsalicylic acid, CAS 50-78-2) in humans include: 1. Gastrointestinal irritation and erosions 2. Tinnitus (ringing in the ears) 3. Decreased platelet aggregation and prolonged bleeding time 4. Metabolic acidosis 5. Respiratory alkalosis (increased respiratory rate) 6. Hyperthermia (in higher doses) 7. Hepatotoxicity (liver damage) 8. Renal impairment (kidney dysfunction) 9. Electrolyte imbalances 10. Hypersensitivity reactions 11. Bronchoconstriction in susceptible individuals (aspirin-exacerbated respiratory disease) 12. Salicylism (headache, dizziness, confusion, sweating) 13. Reduced renal prostaglandin synthesis 14. Increased risk of gastrointestinal bleeding	
GPT-4o	1. Gastrointestinal irritation and bleeding 2. Hepatotoxicity (liver damage) 3. Renal impairment (kidney damage) 4. Tinnitus (ringing in the ears) 5. Dizziness 6. Nausea and vomiting 7. Allergic reactions (e.g., rash, asthma exacerbation) 8. Metabolic acidosis 9. Hyperventilation 10. Increased bleeding time These effects are based on the known pharmacological and toxicological profile of	*[The following response was taken from model's training knowledge]* To address the query regarding the subacute toxicity of Aspirin (CAS number 50-78-2) on humans, I will provide a list of relevant information based on available scientific data and toxicological assessments. Subacute toxicity refers to the effects of a substance following repeated exposure over a period of time, typically ranging from a few days to several weeks. 1. **Gastrointestinal Effects**: Repeated exposure to Aspirin can lead to gastrointestinal irritation, including symptoms such as nausea, vomiting, and gastric bleeding. These effects are due to the inhibition of prostaglandin synthesis, which protects the gastric mucosa. 2. **Renal Impairment**: Aspirin can cause renal impairment with prolonged use, particularly in individuals with pre-existing kidney conditions. This is due to the	** Tools ** ** Toxicity ** Aspirin (CAS 50-78-2) is associated with various toxicological effects, including gastrointestinal issues, bleeding risks, and developmental abnormalities such as cleft palate and atrial septum defect. These effects are documented in the ToxRefDB database. (source: ToxRefDB) ** RAG ** No results were found for the query on sub-acute toxicity of aspirin on humans. (source: RAG search) ** Literature ** A study by Wong et al. (2016) indirectly addresses aspirin's sub-acute toxicity by comparing it to Rh2E2, highlighting aspirin's potential adverse effects like gastrointestinal issues and bleeding risks when used long-term. (source: Wong et al., 2016) ** Training Data ** Aspirin is known to cause gastrointestinal irritation, bleeding, and

Performance comparison

ToxPipe Evaluation

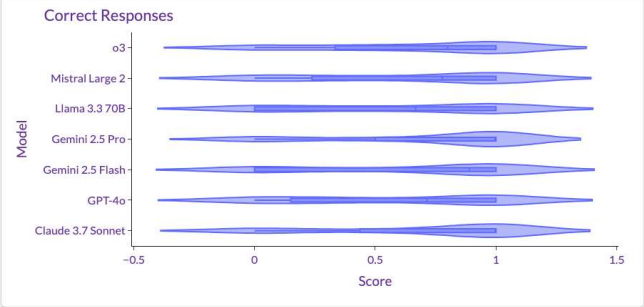
Set Eval | Evals | Compare

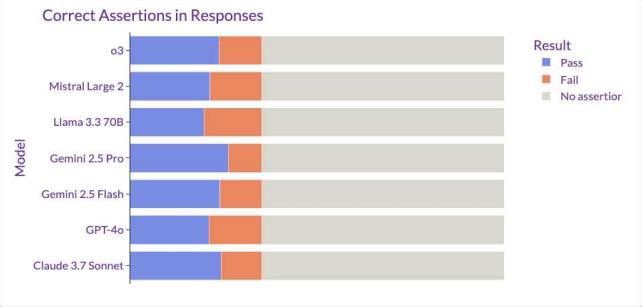
Prompts | Graphs | Table

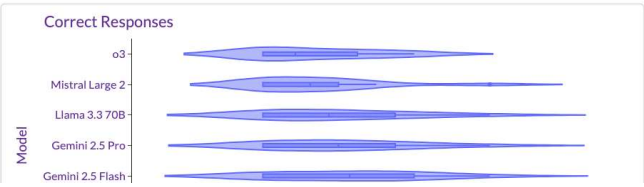
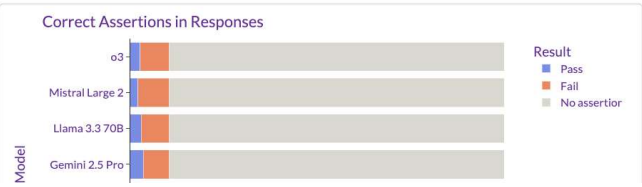
Levels: Any | Eval set: Any | Species: Any | Evals: Any | Hide unlabeled evals

base-model_tox-type-assertion-prompts_human

Effect type: Any | Chemical and CASRN: Any





base-model_tox-type-assertion-prompts_rat

Effect type: Any | Chemical and CASRN: Any